Expt No: 08

Nmap using Try Hack Me

Aim:

To discover live host on a network using Nmap host discovery scans - ARP scan, ICMP scan and TCP/UDP ping scan - and to understand which protocol operates on which layer of TCP/IP model and when each one applies.

ARP host Discovery.

$ sudo nmap -PR -sn 10.200.6.0/24

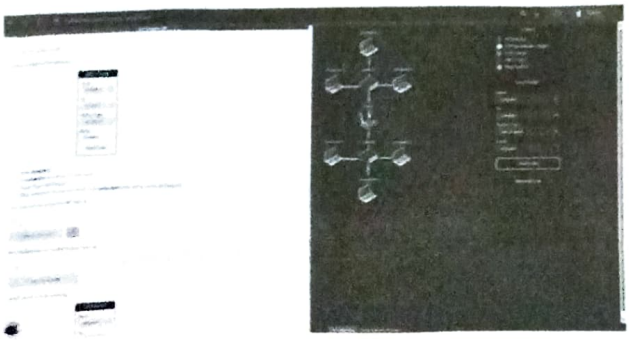
o/p

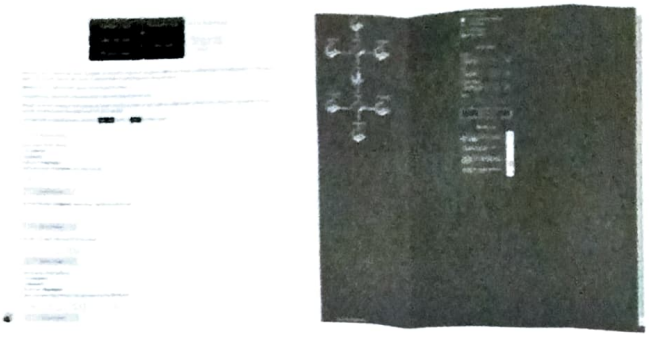
Starting Nmap 7.80 (https://nmap.org)
Nmap scan report for 10.200.6.1
Host is up
Nmap done: 256 IP addres (1 host up)
scanned in 10.51 seconds

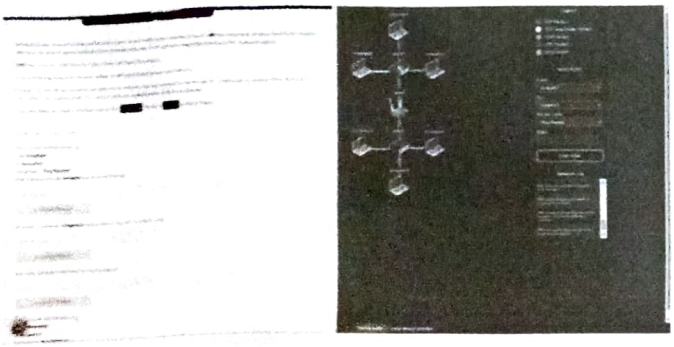
ICMP Host Discovery

ICMP Echo (-PE)

$sudo nmap -PE -sn 10.200.6.0/24
Nmap scan report for 10.200.6.1
Host is up (0.069s latency)
Nmap scan report for 10.200.6.50
Host is up (0.070 latency)
Nmap scan report for 10.200.6.250
Host is up (0.069s latency)
Nmap done: 256 IP addres (3 hosts up)
scanned in 3.38 seconds.

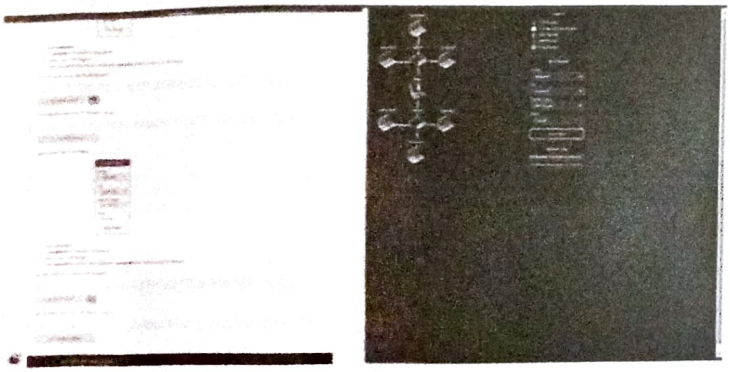


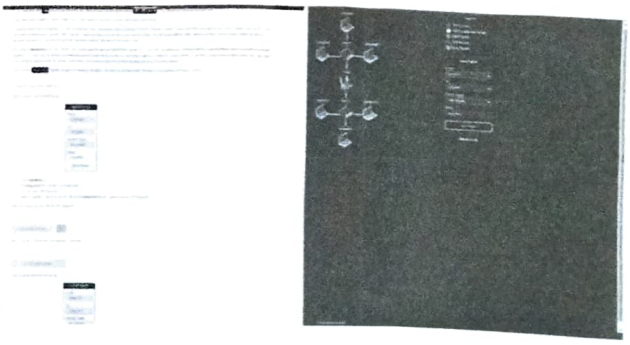




```
pentester@TryHackMe$ sudo nmap -PE -sn 10.10.68.220/24

Starting Nmap 7.92 ( https://nmap.org ) at 2021-09-02 12:16 EEST
Nmap scan report for 10.10.68.50
Host is up (0.12s latency).
Nmap scan report for 10.10.68.52
Host is up (0.12s latency).
Nmap scan report for 10.10.68.77
Host is up (0.11s latency).
Nmap scan report for 10.10.68.110
Host is up (0.11s latency).
Nmap scan report for 10.10.68.140
Host is up (0.11s latency).
Nmap scan report for 10.10.68.142
Host is up (0.11s latency).
Nmap scan report for 10.10.68.220
Host is up (0.11s latency).
Nmap scan report for 10.10.68.222
Host is up (0.11s latency).
Nmap done: 256 IP addresses (8 hosts up) scanned in 8.26 seconds
```

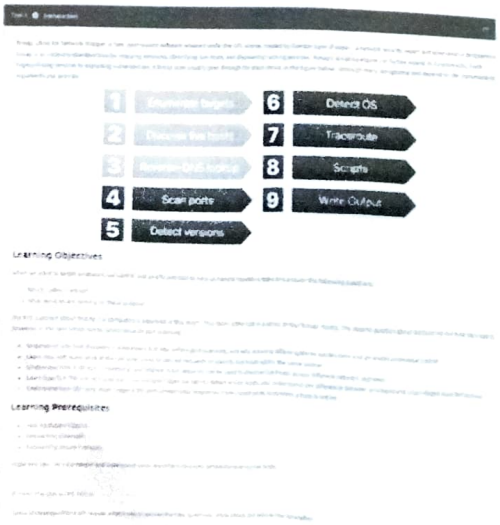




```
pentester@TryHackMe$ sudo nmap -PE -sn 10.10.210.6/24

Starting Nmap 7.60 ( https://nmap.org ) at 2021-09-02 07:12 BST
Nmap scan report for ip-10-10-210-75.eu-west-1.compute.internal (10.10.210.75)
Host is up (0.00013s latency).
MAC Address: 02:03:75:58:F2:89 (Unknown)
Nmap scan report for ip-10-10-210-100.eu-west-1.compute.internal (10.10.210.100)
Host is up (-0.10s latency).
MAC Address: 02:42:D0:10:7D:C9 (Unknown)
Nmap scan report for ip-10-10-210-165.eu-west-1.compute.internal (10.10.210.165)
Host is up (0.00022s latency).
MAC Address: 02:59:F9:4F:17:97 (Unknown)
Nmap scan report for ip-10-10-210-6.eu-west-1.compute.internal (10.10.210.6)
Host is up
Nmap done: 256 IP addresses (4 hosts up) scanned in 3.12 seconds
```

```
pentester@TryHackMe$ sudo nmap -PU -sn 10.10.68.220/24
Starting Nmap 7.92 ( https://nmap.org ) at 2021-09-02 13:45 EEST
Nmap scan report for 10.10.68.52
Host is up (0.10s latency).
Nmap scan report for 10.10.68.121
Host is up (0.10s latency).
Nmap scan report for 10.10.68.125
Host is up (0.14s latency).
Nmap scan report for 10.10.68.134
Host is up (0.096s latency).
Nmap scan report for 10.10.68.220
Host is up (0.11s latency).
Nmap done: 256 IP addresses (5 hosts up) scanned in 9.20 seconds
```



ICMP Timestamp (-PP)

```
$nmap -PP -sn 10.200.6.0/24
Nmap scan report for 10.200.6.50
Host is up (0.069s latency)
Nmap scan report for 10.200.6.250
Host is up (0.069s latency)
Nmap done: 256 IP addres (3 hosts up) scanned
           in 8.44 second
```

ICMP Addres Mask (-PM)

```
$nmap -PM -sn 10.200.6.0/24
Nmap done: 256 IP addres (0 hosts up) scanned in
           52.14 seconds
```

TCP UDP Ping Discovery

i) TCP SYN Ping (-PS)

```
$nmap -PS -sn 10.200.6.0/24
Nmap scan report for 10.200.6.50
Host is up (0.07s latency)
Nmap scan report for 10.200.6.250
```

ii) TCP ACK Ping (-PA)

Sends a TCP packet with the ACK flag set

```
$sudo nmap -PA -sn 10.200.6.0/24
Nmap scan report for 10.200.6.50
Host is up (0.070s latency)
Nmap scan report for 10.200.6.250
Host is up (0.069s latency)
Nmap done: 256 IP addres (2 hosts up) scanned
in 36.64 seconds
```

iii) UDP Ping

```
$ sudo nmap -PU -sn 10.200.6.0/24
Nmap scan report for 10.200.6.50
Host is up (0.070 Latency)
Nmap scan report for 10.200.6.250
Host is up (0.069 latency)
```

Nmap done: 256 IP address (2 hosts up) scanned in 56.97 seconds

Masscan - A Faster Alternative

```
$masscan 10.200.60.0/24 -p443
$masscan 10.200.60.0/24 -p80,443
$masscan 10.200.60.0/24 -p22-25
```

Result:

Thus live host discovery for ARP, ICMP TCP UDP was studied and the output is verified successfully.